DSA 107 · DATA SCIENCE AND ANALYTICS LABORATORY

Lab 3 Report

Groups, and Two Quantitative Variables

Fuel economy by origin, weight and engine size in the Auto MPG data

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1. Objective

This third lab asked how groups differ, and how two quantitative variables move together. The aim was to practise four new tools: boolean masks for counting and filtering rows; groupby for splitting rows into groups and comparing their summaries, including two keys at once; the correlation coefficient as a one-number version of a scatter plot; and reading a box plot, a figure the lecture had not covered. My own test for the session: whenever two groups differ, ask what else is different between them.

2. Data Description

The dataset is Auto MPG: 398 car models sold in the United States between 1970 and 1982, read from the course's teaching copy in the seaborn-data repository directly from its URL with pd.read_csv. Each row is one car model, and the nine columns cover fuel efficiency mpg (miles per gallon, higher is better), engine size, horsepower, weight, acceleration, the model year, the origin of the car - usa, europe or japan - and its name (Table 1).

Nothing was cleaned and all rows were kept. The data is almost complete: only horsepower has missing values, 6 of the 398 rows (about 1.5%). That is small enough to leave in place - the weight-mpg readings below use all 398 cars, and only the correlation involving horsepower uses the 392 cars that have a value. Table 1 collects the details.

Property	Value
Source	seaborn-data copy of Auto MPG (github.com/mwaskom/seaborn-data, mpg.csv); original: UCI Machine Learning Repository
Size	398 rows × 9 columns (one car model per row)
Variable of interest	mpg - fuel efficiency in miles per gallon; higher is better
Key features	numeric: cylinders, displacement, horsepower, weight (lb), acceleration (s), model_year; categorical: origin (usa / europe / japan), name
Missing values	horsepower: 6 rows (1.5%); all other columns complete - none dropped
After preprocessing	no cleaning applied; all 398 rows retained

Table 1. Dataset summary and what was (not) done to it.

3. Methodology

One Colab notebook, run top to bottom after a restart, using pandas 2.2.3 with matplotlib and seaborn (whitegrid). The steps, in order, with the reason behind each:

1. **Load and look.** read_csv(url); .shape shows (398, 9); head(), and isna().sum().sort_values(ascending=False) puts the one incomplete column first.
2. **Boolean masks.** cars["mpg"] >= 30 makes one True/False answer per row; .sum() counts the yes-rows and .mean() turns them into a fraction; cars[cars["origin"] == "japan"] keeps only the matching rows.
3. **Group summaries.** Split-apply-combine: groupby("origin")["mpg"] with .mean() and .count(), checked against len(cars); then two keys at once, origin and cylinders, with .unstack() into a grid.
4. **Two quantitative variables.** A scatter plot first (weight against mpg, alpha=0.5 so overlaps show), then one number: cars[["mpg", "weight", "horsepower"]].corr(), reading the weight-mpg cell with .loc; a second scatter colours the dots by origin.
5. **Box plot.** The ready-made cell was run unchanged; the work here was reading it: the median line, the box as the middle 50%, the whiskers, and outliers as individual dots.
6. **Save.** the finished notebook was pushed to my dsa107-labs repository on GitHub.

4. Results

The masks split the data quickly: 92 cars reach 30 mpg or more, a fraction of 0.231 of all 398, and 79 cars were made in Japan.

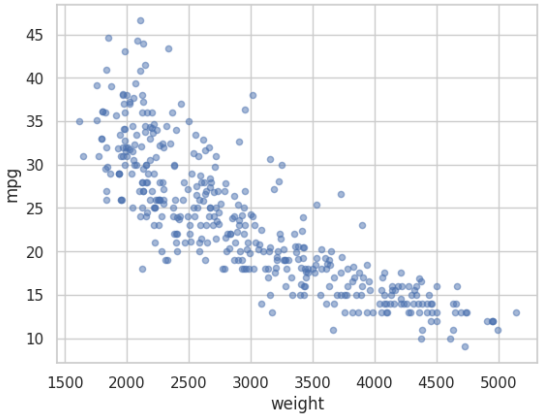


Figure 1. Weight against fuel efficiency, all 398 cars.

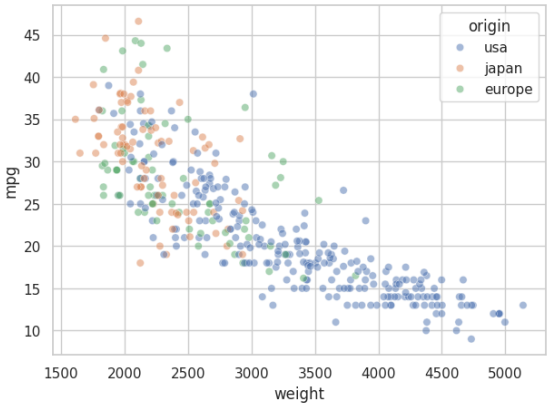


Figure 2. The same cars, coloured by origin.

Figure 1 shows a clear negative pattern - heavier cars travel fewer miles per gallon - and the correlation table turns it into one number: r = -0.83, a strong negative relationship. Figure 2 adds the groups: US cars sit bottom-right (heavier, lower mpg), Japanese and European cars cluster top-left (lighter, higher mpg).

origin	3	4	5	6	8
europe	-	28.4	27.4	20.1	-
japan	20.6	31.6	-	23.9	-
usa	-	27.8	-	19.7	15.0

Table 2. Mean mpg for every origin and cylinder combination, unstacked into a grid (a dash: no cars in that combination).

The group means repeat the pattern: Japan 30.45, Europe 27.89, the USA 20.08, with 79, 70 and 249 cars respectively. Table 2 shows how much of that gap is engine size: among 4-cylinder cars the Japan-USA gap is only 3.8 mpg (31.6 against 27.8) versus 10.4 overall, and the 8-cylinder column belongs to the USA alone.

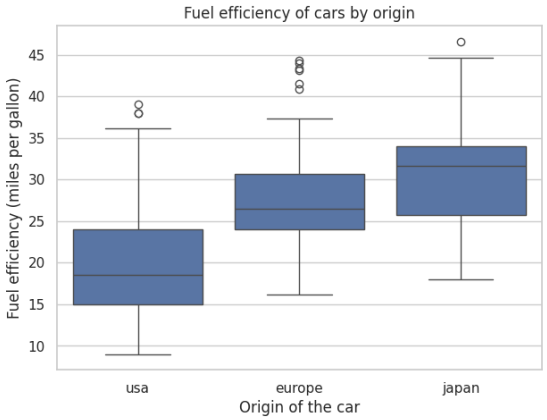


Figure 3. Fuel efficiency by origin as a box plot (ready-made cell, order: usa, europe, japan).

The box plot agrees with the means but shows more. Japan has the highest median (about 31.5 mpg) against the USA's lowest (about 18.5), and Europe has the shortest box - its middle 50%, roughly 24 to 31, sit closest together - with the most outliers, four unusually efficient cars at 41-44 mpg. The overlap matters: Japan's lowest whisker reaches down to about 18 mpg, so not every Japanese car beats every American one (the USA reaches about 38).

5. Discussion

The central lesson is question 3c: "made in the USA" is not itself the explanation for low mpg. The group means show a 10.4 mpg gap, but among 4-cylinder cars it shrinks to 3.8 (Table 2): origin is mixed together with a design habit of the era, because US manufacturers sold far more large-engine cars. The figures add two warnings: a strong correlation (r = -0.83) describes the weight-mpg link but cannot prove its mechanism, and groups can differ at the middle while overlapping at the edges - exactly what the box plot's whiskers show. Model_year is ignored here - a natural next step is mean mpg by model_year.

6. Reflection

Two things were genuinely new. The first was the two-key groupby: mean mpg per origin was immediate, but the origin × cylinders grid needed .unstack() explained twice before it clicked - and one dash in that grid said more than any number, because it meant "no such car exists", not "missing data". The second was the box plot, which I could not read on the first pass; after Gemini walked me through its anatomy I checked every claim against the figure - the four European outliers are real dots, and Japan's low whisker really does reach into American territory. What I understand now: a group difference is only the start - the next question is always what else differs between the groups.

7. References

[1] Seaborn-data teaching copy of the Auto MPG dataset (mpg.csv), github.com/mwaskom/seaborn-data - the file read by the notebook; original distribution: Auto MPG, UCI Machine Learning Repository, archive.ics.uci.edu.

[2] Waskom, M. (2021). seaborn: statistical data visualization. Journal of Open Source Software, 6(60), 3021.

Notebook repository: https://github.com/Anerea/dsa107-labs